Pedram Akbarian

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akbarian@utexas.edu pedakb.github.io github.com/pedakb

Education

The University of Texas at Austin, *Austin, TX*
Ph.D. in Electrical and Computer Engineering (GPA: 4.0/4.0) — May 2027 (Expected)
M.S.E. in Electrical and Computer Engineering — May 2025
Advisors: Prof. Nhat Ho and Prof. Atlas Wang

University of Tehran, *Tehran, Iran*
B.Sc. in Electrical Engineering (*Minor* in Computer Engineering) — May 2019
Thesis: "Sparse Subspace Clustering (SSC): Applications in Human Motion Segmentation"
Advisor: Prof. Babak N. Araabi

Research Interests

- **Stable & Efficient Training for Deep Learning:** Studying optimization stability and statistical efficiency that improve the scalability and robustness of large foundation models, with a focus on sparse architectures.
- **Foundation Models for Unconventional Domains:** Focusing on stable and efficient training of deep models to enable end-to-end learning on unstructured and noisy data (e.g., time series, spatio-temporal data).

Honors and Awards

- **Silver Medal** recipient in the 26th Iranian National Physics Olympiad — Sep. 2013
- **Bronze Medal** recipient in the 25th Iranian National Physics Olympiad — Sep. 2012
- Recipient of the **Grant** from the **National Elites Foundation**, for Silver and Bronze Medals of National Physics Olympiad and outstanding academic success — Nov. 2014 – Jun. 2019

Work Experience

Quantitative Research Intern, *Point72 Asset Management, New York, NY* — Summer 2026
- Quantitative Research Team, Internal Alpha Capture (IAC).
- Researched **neural network learning dynamics** for U.S. equity models.
- Developed analysis tools to monitor learning dynamics across different market regimes and subuniverses.
- Designed and evaluated alternative **loss functions** tailored to the signal pipeline and analyzed their tradeoffs.

AI Research Intern, *Toyota InfoTech Lab, Mountain View, CA* — Jan. 2024 – May 2025
- Developed **time-series foundation models** with a focus on scalability and efficiency.
- Integrated **Mixture of Experts (MoE)** into transformer and non-transformer forecasting architectures.

Data Scientist Intern, *H-E-B, San Antonio, TX* — Summer 2022
- Developed an online optimization algorithm for **order batching** under time and space constraints.

AI Scientist Intern, *CognitiveScale, Austin, TX* — Summer 2021
- Developed methods for **counterfactual explanations** for time-series data.

Relevant Graduate Courses (Graduate courses taken during undergraduate studies are indicated by †)

- **Information Theory** Spring 2022
- **Stochastic Control Theory** Spring 2021
- **Combinatorial Optimization** Fall 2020
- **Large Scale Optimization II** Spring 2020
- **Probabil. & Stochastic Procs.** Fall 2019
- **Stochastic Processes**† Fall 2018
- **Online Learning** Fall 2021
- **Advanced Probability** Fall 2020
- **Theoretical Statistics** Spring 2020
- **Statistical Machine Learning** Spring 2020
- **Convex Optimization** Fall 2019
- **Pattern Recognition**† Fall 2017

Publications

[8] Fanqi Yan, Huy Nguyen, Dung Le, **Pedram Akbarian**, Nhat Ho, Alessandro Rinaldo. *On Minimax Estimation of Parameters in Softmax-Contaminated Mixture of Experts.* The Thirty-ninth Annual Conference on Neural Information Processing Systems **(NeurIPS)**, 2025.

[7] Fanqi Yan, Huy Nguyen, Dung Le, **Pedram Akbarian**, Nhat Ho. *Understanding Expert Structures on Minimax Parameter Estimation in Contaminated Mixture of Experts.* The 28th International Conference on Artificial Intelligence and Statistics **(AISTATS)**, 2025.

[6] Huy Nguyen, **Pedram Akbarian**[*], Trang Pham[*], Trang Nguyen, Shujian Zhang, Nhat Ho. *Statistical Advantages of Perturbing Cosine Router in Sparse Mixture of Experts.* The Thirteenth International Conference on Learning Representations **(ICLR)**, 2025.

[5] Huy Nguyen, **Pedram Akbarian**, Nhat Ho. *Is Temperature Sample Efficient for Softmax Gaussian Mixture of Experts?* Proceedings of the International Conference on Machine Learning **(ICML)**, 2024.

[4] Huy Nguyen, **Pedram Akbarian**, Trungtin Nguyen, Nhat Ho. *A General Theory for Softmax Gating Multinomial Logistic Mixture of Experts.* Proceedings of the International Conference on Machine Learning **(ICML)**, 2024.

[3] **Pedram Akbarian**[*], Tongzheng Ren[*], Jiacheng Zhuo, Sujay Sanghavi, Nhat Ho. *Improving Computational Complexity in Statistical Models with Local Curvature Information.* Proceedings of the International Conference on Machine Learning **(ICML)**, 2024.

[2] Huy Nguyen, **Pedram Akbarian**, Fanqi Yan, Nhat Ho. *Statistical Perspective of Top-K Sparse Softmax Gating Mixture of Experts.* The Twelfth International Conference on Learning Representations **(ICLR)**, 2024.

[1] Tina Han, Jette Henderson, **Pedram Akbarian**, Joydeep Ghosh. *Improving Counterfactual Explanations for Time Series Classification Models in Healthcare Settings.* NeurIPS Workshop on Learning from Time Series for Health, 2022.

Preprints

[2] Fanqi Yan, Huy Nguyen, **Pedram Akbarian**, Nhat Ho, Alessandro Rinaldo. *Sigmoid Self-Attention is Better than Softmax Self-Attention: A Mixture-of-Experts Perspective.* arXiv:2502.00281, 2025.

[1] **Pedram Akbarian**[*], Huy Nguyen[*], Xing Han[*], Nhat Ho. *Quadratic Gating Mixture of Experts: Statistical Insights into Self-Attention.* arXiv:2410.11222, 2024.

(* denotes equal contribution.)

Selected Teaching Experience (Graduate courses are indicated by †)

Graduate Teaching Assistant, *The University of Texas at Austin*

- ◇ **Probability & Stochastic Processes**†
- ◇ **Probability/Random Processes**
- ◇ **Digital Signal Processing**
- ◇ **Statistical Machine Learning**†
- ◇ **Data Science Principles**
- ◇ **Data Science Lab**

Undergraduate Teaching Assistant, *University of Tehran*

- ◇ **Pattern Recognition**†
- ◇ **Statistical Inference**†

Professional Services

- ◇ Reviewer at the Conference on Neural Information Processing Systems (NeurIPS) 2024-2026.
- ◇ Reviewer at the International Conference on Learning Representations (ICLR) 2025.
- ◇ Reviewer at the Association for the Advancement of Artificial Intelligence (AAAI) 2025.
- ◇ Reviewer at the International Conference on Artificial Intelligence and Statistics (AISTATS) 2024-2025.

Skills

- ◇ **Programming Languages:** Python (*proficient*), C/C++, SQL, MATLAB, R, LaTeX
- ◇ **Software and Frameworks:** PyTorch (*proficient*), TensorFlow, Hugging Face, Triton, Git

References

Available upon request.